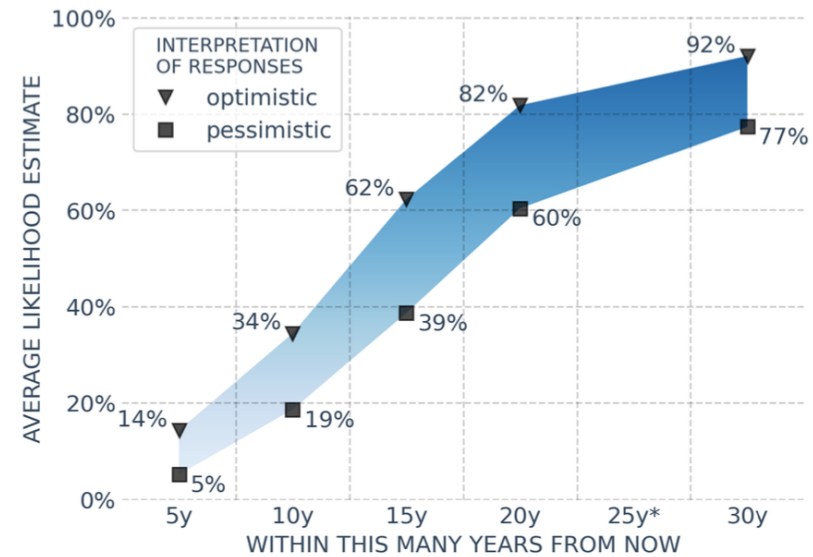
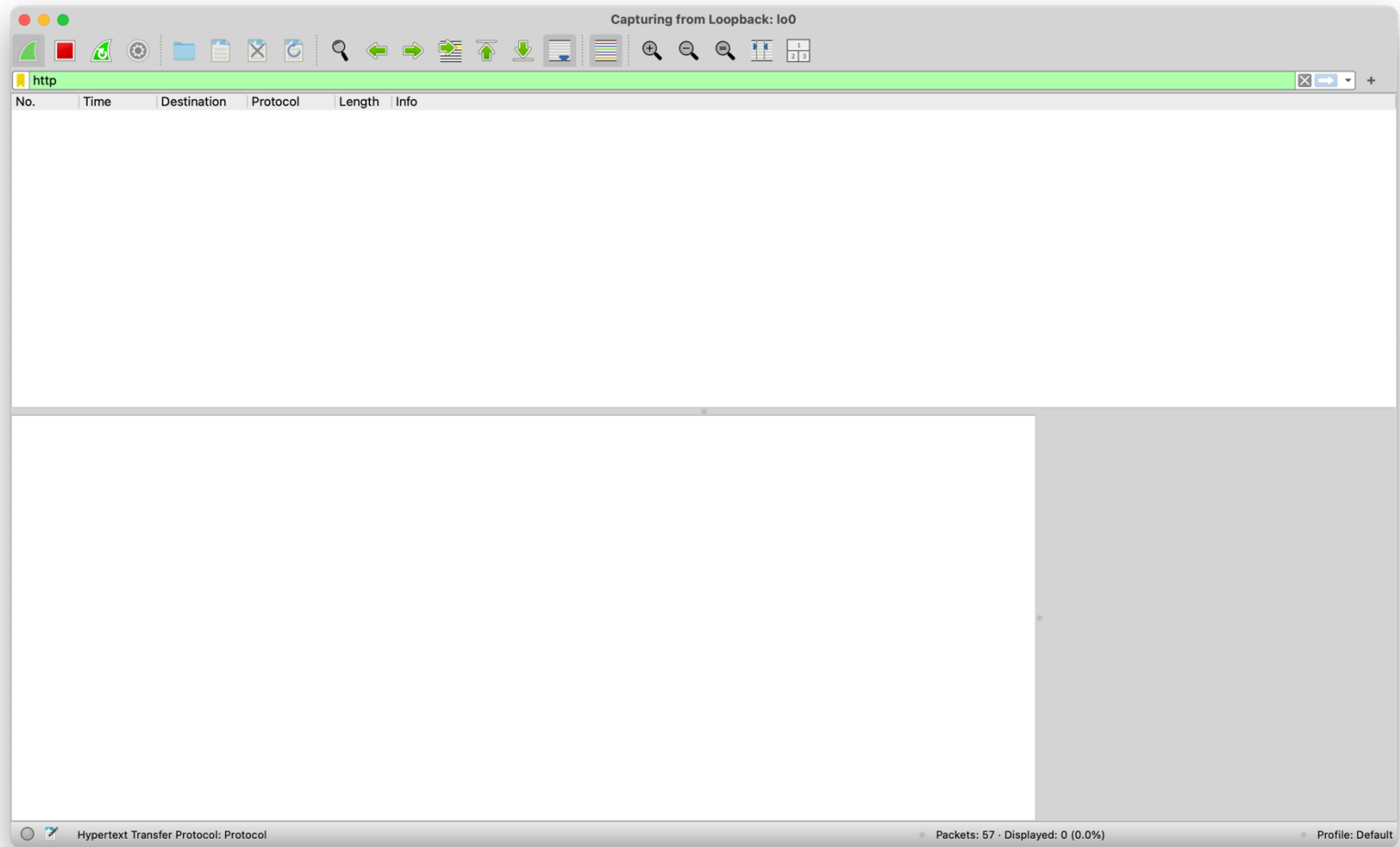


 **2024 OPINION-BASED ESTIMATES OF THE  
LIKELIHOOD OF A QUANTUM COMPUTER  
ABLE TO BREAK RSA-2048 IN 24 HOURS,  
AS FUNCTION OF TIME**



Average likelihood estimates for the realization of a CRQC in the coming years, based on expert opinions. Range between average of an optimistic (top value) or pessimistic (bottom value) of likelihood intervals indicated by the respondents.  
\*The 25 years timeframe was not considered explicitly.



Capturing from Loopback: lo0

http

| No.  | Time        | Destination | Protocol  | Length | Info                                                                                                                                               |
|------|-------------|-------------|-----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1126 | 102.1402... | 127.0.0.1   | HTTP/JSON | 369    | POST /hub/hank/dske/api/v1/key-share HTTP/1.1, JSON (application/json)                                                                             |
| 1126 | 102.1404... | 127.0.0.1   | HTTP/JSON | 369    | POST /hub/helen/dske/api/v1/key-share HTTP/1.1, JSON (application/json)                                                                            |
| 1128 | 102.1404... | 127.0.0.1   | HTTP/JSON | 369    | POST /hub/hilary/dske/api/v1/key-share HTTP/1.1, JSON (application/json)                                                                           |
| 1134 | 102.1407... | 127.0.0.1   | HTTP/JSON | 369    | POST /hub/holly/dske/api/v1/key-share HTTP/1.1, JSON (application/json)                                                                            |
| 1136 | 102.1408... | 127.0.0.1   | HTTP/JSON | 369    | POST /hub/hugo/dske/api/v1/key-share HTTP/1.1, JSON (application/json)                                                                             |
| 1138 | 102.1439... | 127.0.0.1   | HTTP/JSON | 288    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1139 | 102.1439... | 127.0.0.1   | HTTP/JSON | 288    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1142 | 102.1440... | 127.0.0.1   | HTTP/JSON | 288    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1144 | 102.1445... | 127.0.0.1   | HTTP/JSON | 288    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1145 | 102.1445... | 127.0.0.1   | HTTP/JSON | 288    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1168 | 102.1488... | 127.0.0.1   | HTTP/JSON | 272    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1178 | 102.1591... | 127.0.0.1   | HTTP      | 307    | GET /client/connie/etsi/api/v1/keys/sam/dec_keys?key_ID=39b64dd0-c22a-4f2a-b5a0-6061d399d311 HTTP/1.1                                              |
| 1200 | 102.1641... | 127.0.0.1   | HTTP      | 432    | GET /hub/hank/dske/api/v1/key-share?client_name=connie&master_sae_id=sam&slave_sae_id=sofia&key_id=39b64dd0-c22a-4f2a-b5a0-6061d399d311 HTTP/1.1   |
| 1202 | 102.1643... | 127.0.0.1   | HTTP      | 433    | GET /hub/helen/dske/api/v1/key-share?client_name=connie&master_sae_id=sam&slave_sae_id=sofia&key_id=39b64dd0-c22a-4f2a-b5a0-6061d399d311 HTTP/1.1  |
| 1204 | 102.1645... | 127.0.0.1   | HTTP      | 434    | GET /hub/hilary/dske/api/v1/key-share?client_name=connie&master_sae_id=sam&slave_sae_id=sofia&key_id=39b64dd0-c22a-4f2a-b5a0-6061d399d311 HTTP/1.1 |
| 1206 | 102.1648... | 127.0.0.1   | HTTP      | 433    | GET /hub/holly/dske/api/v1/key-share?client_name=connie&master_sae_id=sam&slave_sae_id=sofia&key_id=39b64dd0-c22a-4f2a-b5a0-6061d399d311 HTTP/1.1  |
| 1208 | 102.1649... | 127.0.0.1   | HTTP      | 432    | GET /hub/hugo/dske/api/v1/key-share?client_name=connie&master_sae_id=sam&slave_sae_id=sofia&key_id=39b64dd0-c22a-4f2a-b5a0-6061d399d311 HTTP/1.1   |
| 1210 | 102.1660... | 127.0.0.1   | HTTP/JSON | 473    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1212 | 102.1661... | 127.0.0.1   | HTTP/JSON | 473    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1214 | 102.1665... | 127.0.0.1   | HTTP/JSON | 473    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1216 | 102.1666... | 127.0.0.1   | HTTP/JSON | 473    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |
| 1218 | 102.1668... | 127.0.0.1   | HTTP/JSON | 473    | HTTP/1.1 200 OK, JSON (application/json)                                                                                                           |

Frame 1126: 369 bytes on wire (2952 bits), 369 bytes captured (2952 bits) on interface lo0, id 0

- > Null/Loopback
- > Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- > Transmission Control Protocol, Src Port: 57143, Dst Port: 8102, Seq: 334, Ack: 1, Len: 313
- > [2 Reassembled TCP Segments (646 bytes): #1122(333), #1128(313)]
- > Hypertext Transfer Protocol
  - > POST /hub/hilary/dske/api/v1/key-share HTTP/1.1\r\n
  - Host: 127.0.0.1:8102\r\n
  - Accept: \*/\*\r\n
  - Accept-Encoding: gzip, deflate\r\n
  - Connection: keep-alive\r\n
  - User-Agent: python-httpx/0.28.1\r\n
  - Content-Length: 313\r\n
  - [Content length: 313]
  - Content-Type: application/json\r\n
  - DSKE-Signature: cca3b80c-2e32-49ef-94bc-a257710d23bd:16:32%TIG81Hu0CSZWLNgtz40qw068XbIaiH0PxUln7dZcPts=\r\n\r\n
  - [Response in frame: 1139]

0000 02 00 00 00 45 00 01 6d 00 00 40 00 4  
0010 7f 00 00 01 7f 00 00 01 df 37 1f a6 b  
0020 16 9c f8 1e 00 18 18 eb ff 61 00 00 0  
0030 3d c8 75 98 df 20 21 18 7b 22 6d 61 7  
0040 5f 63 6c 69 65 6e 74 5f 6e 61 6d 65 2  
0050 61 72 6f 6c 22 2c 22 6d 61 73 74 65 7  
0060 65 5f 69 64 22 3a 22 73 61 6d 22 2c 2  
0070 76 65 5f 73 61 65 5f 69 64 22 3a 22 7  
0080 61 22 2c 22 75 73 65 72 5f 6b 65 79 5  
0090 3a 22 33 39 62 36 34 64 64 30 2d 63 3  
00a0 34 66 32 61 2d 62 35 61 30 2d 36 30 3  
00b0 39 39 64 33 31 31 22 2c 22 73 68 61 7  
00c0 6e 64 65 78 22 3a 32 2c 22 65 6e 63 7  
00d0 69 6f 6e 5f 6b 65 79 5f 61 6c 6c 6f 6  
00e0 6f 6e 22 3a 7b 22 66 72 61 67 6d 65 6  
00f0 3a 5b 7b 22 62 6c 6f 63 6b 5f 75 75 6  
0100 22 63 63 61 33 62 38 30 63 2d 32 65 3  
0110 39 65 66 2d 39 34 62 63 2d 61 32 35 3  
0120 64 32 33 62 64 22 2c 22 73 74 61 72 7

Frame (369 bytes) Reassembled TCP (646 bytes)

Packets: 1242 · Displayed: 178 (14.3%) Profile: Default

FastAPI - Swagger UI

+

← → ↺ 🏠

http://localhost:8100/docs

☆ ⏏ ASP New 📄 🗑️ 👤 ⋮

FastAPI

0.1.0 OAS 3.1

/openapi.json

default

^

PUT

/hub/hank/dske/oob/v1/registration

Put Oob Client Registration

▼

GET

/hub/hank/dske/oob/v1/psrd

Get Oob Psrd

▼

POST

/hub/hank/dske/api/v1/key-share

Post Key Share

▼

GET

/hub/hank/dske/api/v1/key-share

Get Key Share

▼

GET

/hub/hank/mgmt/v1/status

Get Mgmt Status

▼

POST

/hub/hank/mgmt/v1/stop

Post Mgmt Stop

▼

Schemas

^

APIAllocation >

Expand all

object

APIBlock >

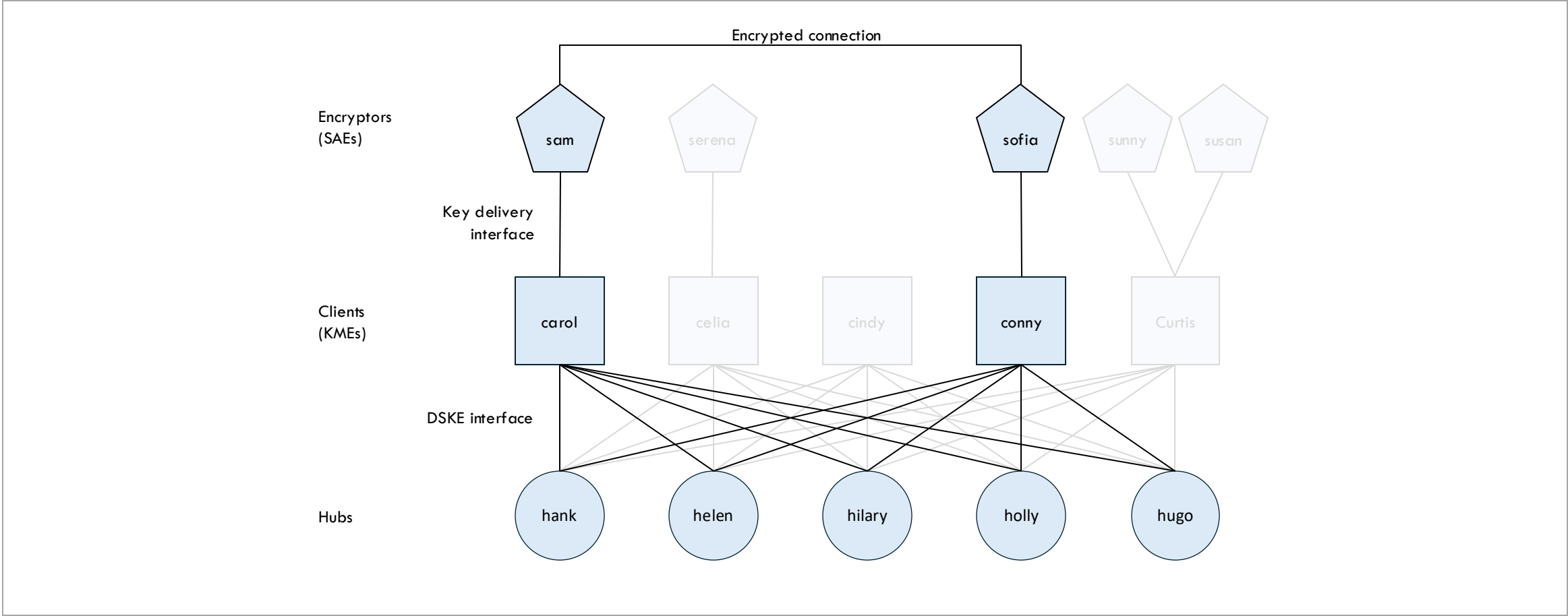
Expand all

object

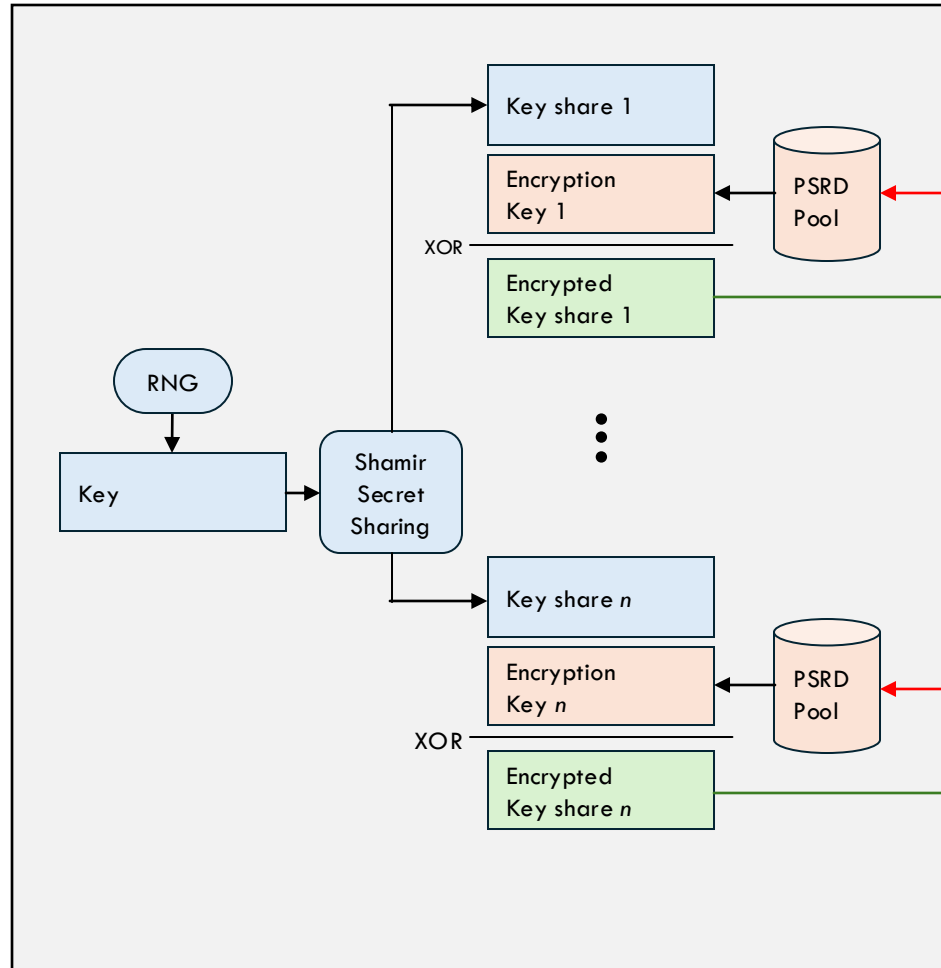
APIEndpoint >

Expand all

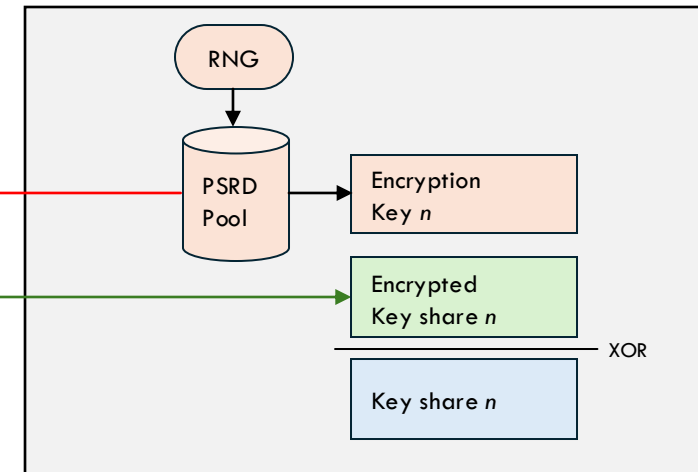
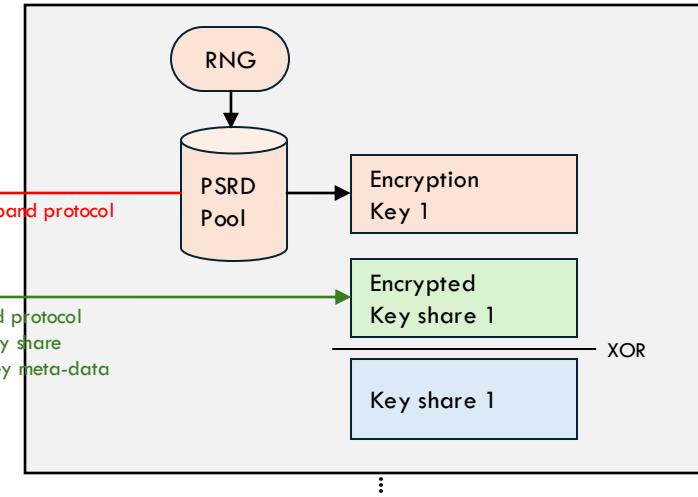
object



Client Carol

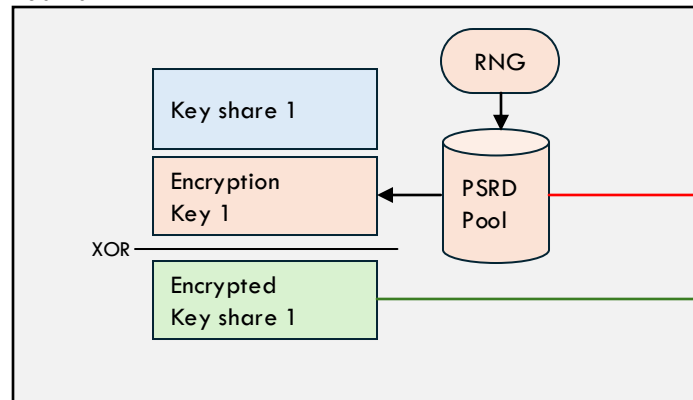


Hub Hank

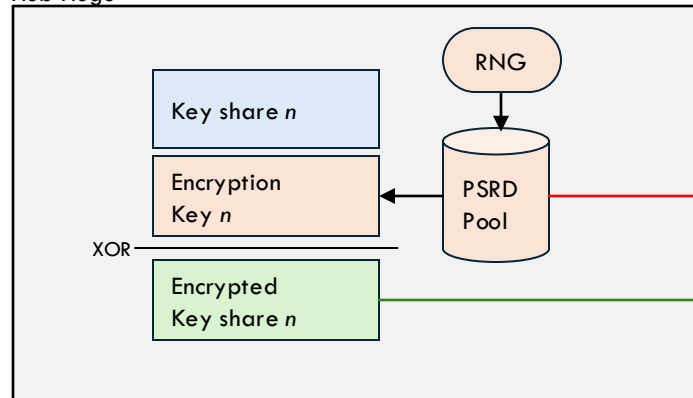


Hub Hugo

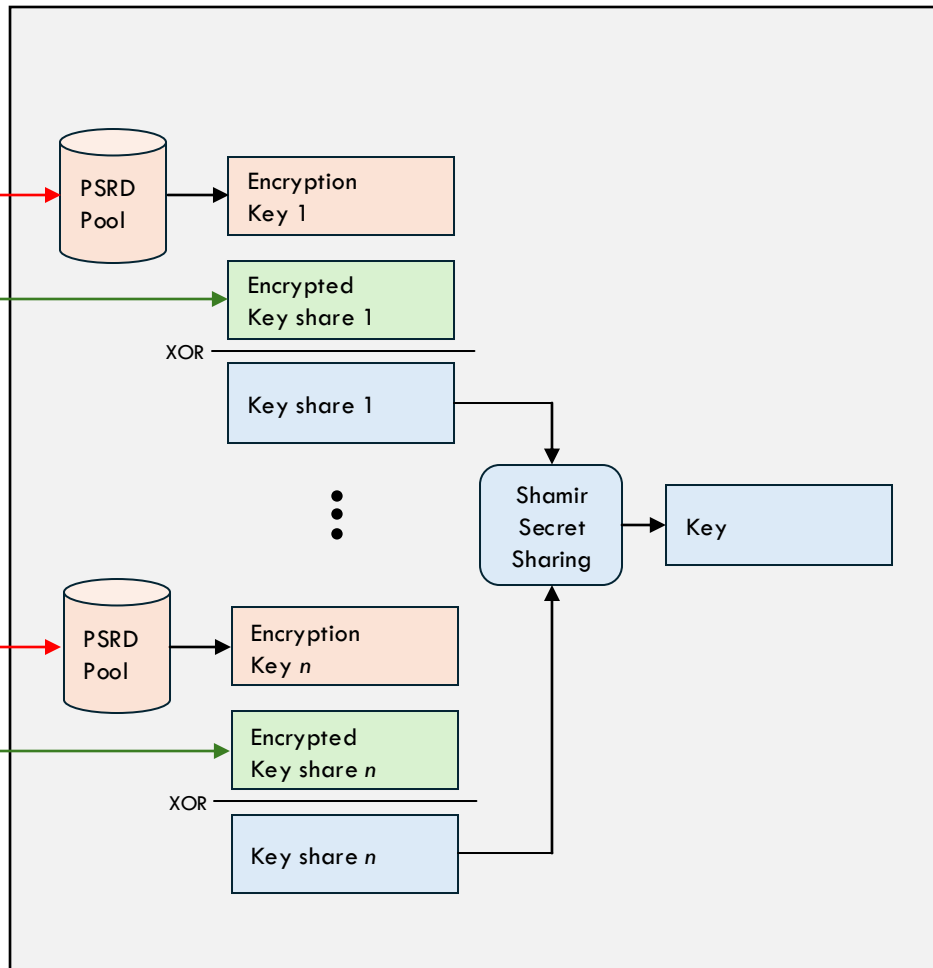
Hub Hank

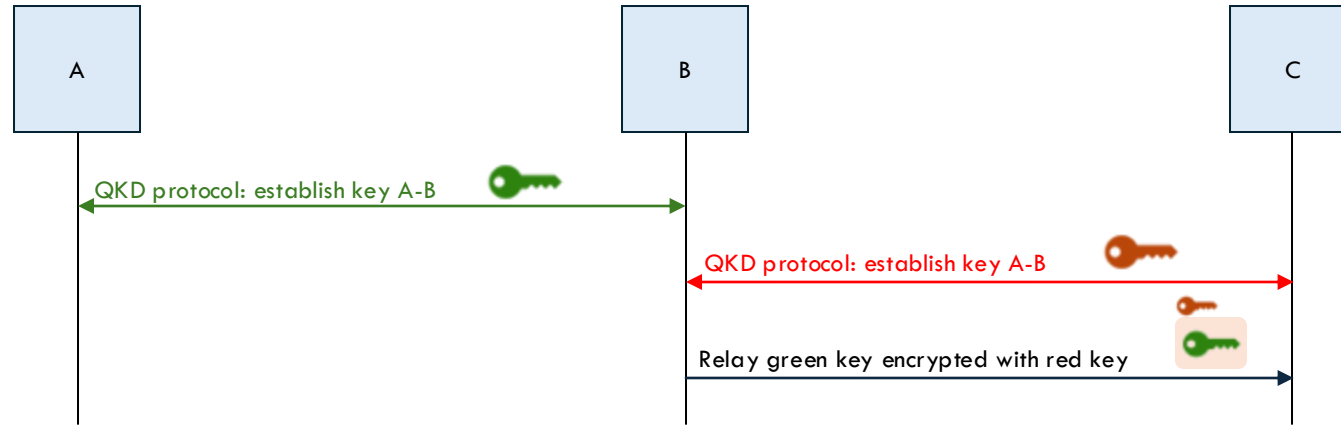


Hub Hugo



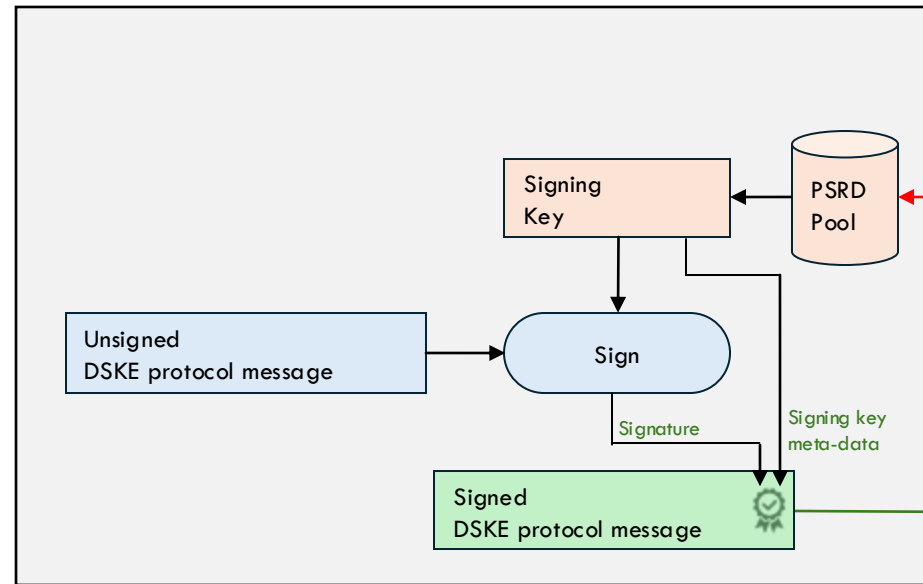
Client Conny



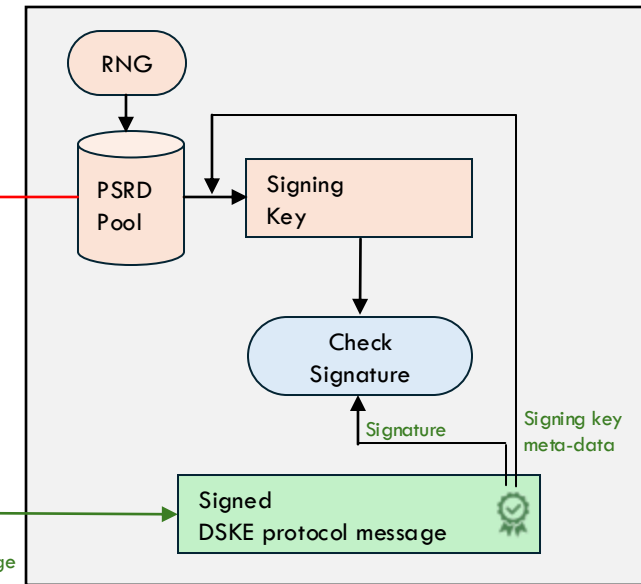




Client Carol



Hub Hank



DSKE out-of-band protocol  
PSRD block

DSKE in-band protocol  
Signed DSKE protocol message  
Signing key meta-data

